Alexander Detkov

Ph.D. Researcher in Computation and Neural Systems, Caltech

adetkov@caltech.edu | alexanderdetkov.github.io | +1 (626) 399 5770

Research on how LLMs and world models store, update and propagate knowledge: structured representations, constraint propagation and continual learning.

Education

California Institute of Technology Sep 2024 – Present
Ph.D. in Computation and Neural Systems, Minor in Mathematics
Advisor: Prof. Matt Thomson | Chen Graduate Fellow | GPA: 4.0/4.0

University of Alberta Sep 2019 – Apr 2024
B.Sc. in Engineering Physics, Minor in Mathematics, with Distinction
Ranked 1st of ∼5,500 graduating undergraduates (C.D. Howe Memorial Fellowship) | GPA: 4.0/4.0

Selected Publications

Do World Models Learn Global Understanding? · Paper
A. Detkov, M. Thomson. *NeurIPS 2026 Workshop on Symmetry and Geometry in Neural Representations; under review at ICLR 2027.*

ImplicitHop: Evaluating Knowledge Integration across Context and Memory · Paper
A. Detkov, D. Karish, M. Oleynikov, M. Thomson. *arXiv preprint, 2026.*

Prompt Baking: LLM Context to Weight Consolidation · Paper Code
A. Detkov[*], A. Bhargava[*], C. Witkowski[*], M. Thomson. *arXiv preprint, 2024.* [*]*Equal contribution.*

Reparameterization through Spatial Gradient Scaling · Paper Code
A. Detkov, M. Salameh, M. Qharabagh, J. Zhang, W. Lui, S. Jui, D. Niu. *ICLR 2023.*

Research Experience

Ph.D. Research — LLMs and World Models Sep 2024 – Present
California Institute of Technology | Prof. Matt Thomson

Learning Global Constraints from Local Observations

- Defined a monoid framework to evaluate whether deep learning models infer global action constraints from local observations or simply learn surface statistics. Applied persistent homology and geometric analysis to model representations identifying linear Fourier structure associated with algebraic constraints.
- Developed compositional training that learns and propagates constraints (96% generalization on inverse, commutativity and composition) while next-state training fails across architectures, constraints and environments.
- Extended the framework to semantic constraints in LLMs and geometric constraints in world models. Traditional training fails to propagate constraints yielding inconsistencies (e.g. reversal curse). Compositional training improves coherence, propagating semantics of relational knowledge in LLMs and geometry in world models.
- Analyzed knowledge propagation formally by defining a constraint proof system where propagation distance is given by minimum proof depth. Showed that transformers poorly propagate constraints deeply with generalization declining sharply with depth, but training on longer action compositions improves deeper propagation.

Prompt Baking: Memory Consolidation of Context to Weights

- Developed a self-distillation approach to consolidate the context of an LLM into LoRA weight updates using a KL-divergence objective. Co-designed applications for instruction following, reasoning and knowledge updates.
- Showed that baking personas and instructions maintain consistent performance across 32-turn conversations while prompting's final performance degrades by 58%. Co-developed half-baking for continuous fine-grained control over personas and behaviors exhibited by the baked LLM.
- Developed Prompt Pursuit, which iteratively consolidates a model's prompted behavior into its weights, improving instruction following by 20% on average over prompting alone.

Research Experience (continued)

Interpretable Neuro-symbolic AI May 2023 – Sep 2023
University of Alberta | Prof. Di Niu

- Developed neuro-symbolic graph neural networks (GNNs) where models are trained iteratively: interpretable logical expressions are distilled from neural components using symbolic regression and used as features for the next iteration of neural training improving out-of-distribution performance.
- Implemented policy-gradient reinforcement learning to learn graph augmentations that regularize GNN training improving model robustness.

Industry Research — Neural Architecture Search and Optimization Jan 2022 – Dec 2022
Huawei Research | Prof. Di Niu and Dr. Mohammad Salameh

- Initiated and led research on the connection between architecture changes in convolutional neural networks (CNNs) and induced changes in training dynamics leading to a first-author ICLR 2023 paper.
- Proved equivalence between branched reparameterization and spatial scaling of gradients in the base network, enabling improved training dynamics without the additional computational cost of branching.
- Developed an information-theoretic method for selecting gradient scalings; evaluated vision models across classification, semantic segmentation and pose estimation.
- Designed hardware-aware architecture search and graph-level optimization of transformer language and vision models for Ascend neural processing units (NPUs).

Earlier Research

Computational Nanofluidics May 2021 – Dec 2021
University of Alberta | Prof. Wylie Stroberg

- Developed and analyzed molecular-dynamics simulations discovering polymer configurations with high absorption. Co-created a Langevin-dynamics simulator for polymer imbibition at longer time and length scales.
- Applied stochastic-process theory to derive mean first-passage times for polymer entry into nanocapillaries; oral presentation at CSME 2022.

Experimental Particle Physics May 2021 – Aug 2021
IceCube Observatory | Prof. Juan Pablo Yáñez

- Applied Bayesian methods towards reconstruction of atmospheric-muon tracks and optical-module calibration.

Honors and Awards

Chen Graduate Fellowship, Caltech Chen Institute for Neuroscience	2024
Rt. Hon. C.D. Howe Memorial Fellowship — highest overall standing of all graduating undergraduates	2024
Governor General's Silver Medal — one of three awarded university-wide	2024
APEGA Past Presidents' Medal — highest standing in Engineering Physics	2024
Dean's Medal in Engineering	2024
NSERC Undergraduate Student Research Award	2023
Faculty of Engineering Dean's Research Award	2021
Undergraduate scholarships — Seven Wardens, Charyk, Enbridge, Iron Standard, McKinney (2×)	2019–24

Technical Skills

ML & LLMs: PyTorch, Hugging Face (Transformers, PEFT, TRL), vLLM.
Model analysis: persistent homology, manifold learning (UMAP, diffusion maps), Fourier and group-representation analysis, probing, steering vectors, sparse autoencoders.
Training & evaluation: LLM post-training and distillation (LoRA, context distillation, SFT, RL fine-tuning); evaluation of reasoning and coding.
Mathematics: semigroup and group theory, automata theory, algebraic topology, differential geometry, dynamical systems, control theory, information theory, statistical mechanics.
Engineering: Python, C++, Julia; Linux, Git, multi-GPU training (DDP).
Languages: English (native), French, Ukrainian, Russian.